

PARTIAL DIFFERENTIAL EQUATIONS

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Heat equation

- Fundamental PDE modeling heat flow, and gives the temperature distribution, $u(x, y, z, t)$ in a body of homogeneous material in space

$$\frac{\partial u}{\partial t} = c^2 \nabla^2 u, \quad c^2 = K/\rho C$$

c^2 is the thermal diffusivity. K Is the thermal conductivity, C the specific heat, and ρ the density of the material in the body.

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}$$

- The heat equation is also called the diffusion equation because it also models chemical diffusion processes.

One – Dimensional Heat Equation

1 – D Heat equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$u(0, t) = 0, \quad u(L, t) = 0, \quad \text{for all } t \geq 0$$



Bar under consideration

Boundary conditions

$$u(x, 0) = f(x), \quad (0 \leq x \leq L) \quad \text{Initial conditions}$$

Solution Procedure

- Method of separating variables, or product method

$$u(x, t) = F(x)G(t)$$



$$u_n(x, t) = B_n \sin \frac{n\pi x}{L} e^{-\lambda_n^2 t}, \quad n = 1, 2, \dots$$

- These functions are called the eigenfunctions, or characteristic functions, and the values $\lambda_n = cn\pi/L$ are called the eigenvalues, or characteristic values.

One-Dimensional Heat Equation

Solution: Fourier series

$$u(x, t) = \sum_{n=1}^{\infty} u_n(x, t) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L} e^{-\lambda_n^2 t}$$

Steady 2D Heat Problems – Laplace's Equation

Laplace's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

For steady – state equilibrium problems governed by a Laplace's equation, three types of boundary conditions can be imposed:

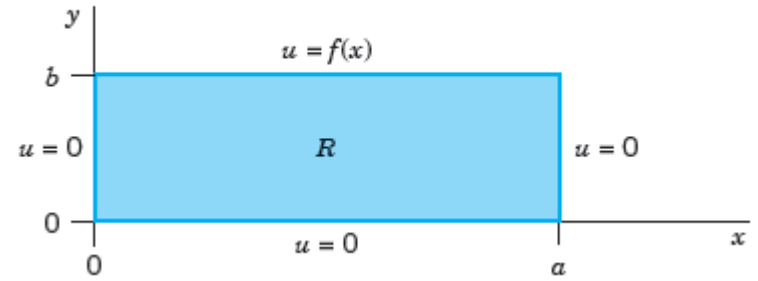
- Dirichlet boundary conditions: The value of the function is specified
 u is specified at the boundary
- Neumann boundary condition: The value of the derivative normal to the boundary is specified
 $\frac{\partial u}{\partial n}$ is specified at the boundary
- Mixed boundary condition: A combination of the function, and its normal derivative is specified on the boundary

$$au + b \frac{\partial u}{\partial n} \text{ is specified at the boundary}$$

Solution Procedure

- Method of separating variables, or product method

$$u(x, y) = F(x)G(y)$$



Rectangle R and given boundary values

Heat Equation: Modeling Very Long Bars

1 – D Heat equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$u(x, 0) = f(x), \quad (-\infty \leq x \leq \infty)$$

Solution Procedure

- Method of separating variables, or product method

$$u(x, t) = F(x)G(t)$$

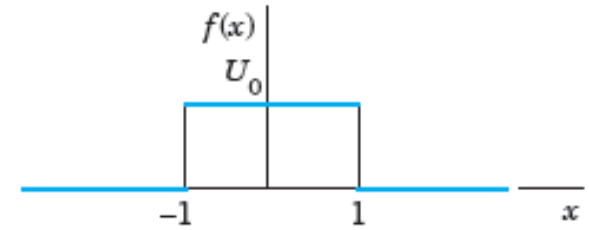
Sample Problem

One-dimensional heat conduction

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad x \in (-\infty, \infty)$$

Initial conditions

$$f(x) = \begin{cases} U_0, & \text{if } |x| < 1 \\ 0, & \text{if } |x| > 1 \end{cases}$$

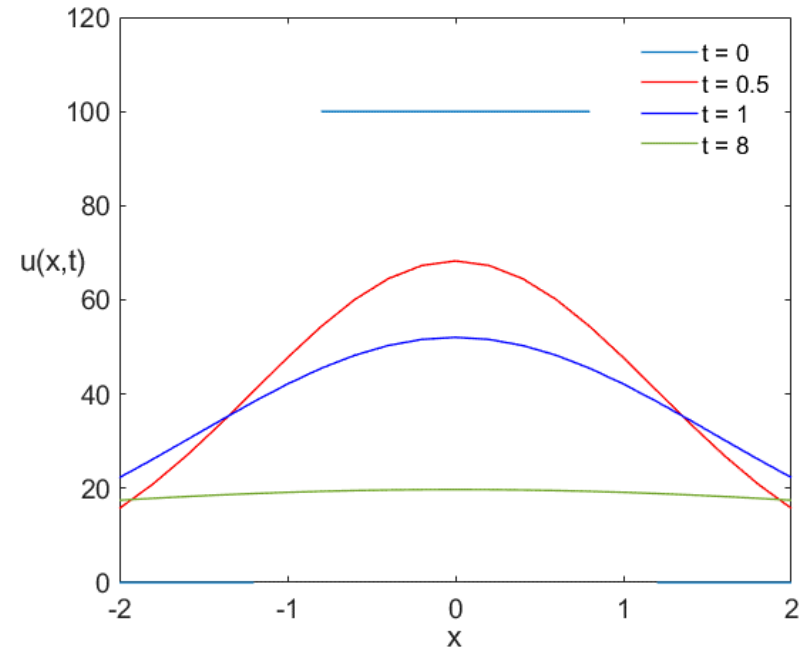


Initial Temperature distribution
in the infinite bar

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t = 0.5; % Given time, t
x = -2:0.2:2;
U0 = 100;
n = size(x,2);
u = zeros(1,n);
% Function u(x,t) calculation
syms z
f = exp(-z^2);
for i = 1:n
    y1 = -(1 + x(i))/(2*sqrt(t));
    y2 = (1 - x(i))/(2*sqrt(t));
    u(i) = double(int(f,y1,y2));
end
% Plot the result
plot(x, (U0/sqrt(pi))*u)

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Solution $u(x,t)$ for several values of t

Problem 1: Find the temperature $u(x, t)$ in a laterally insulated copper bar 80 cm long if the initial temperature is $100 \sin\left(\frac{\pi x}{80}\right)^\circ\text{C}$ and the ends are kept at 0°C . How long will it take for the maximum temperature in the bar to drop to 50°C . Data: $\rho = 8.92 \frac{\text{g}}{\text{cm}^3}$, $c_p = 0.092 \text{ cal}/(\text{g } ^\circ\text{C})$ and $\kappa = 0.95 \text{ cal}/(\text{cm sec } ^\circ\text{C})$

Problem 2: One-dimensional heat conduction

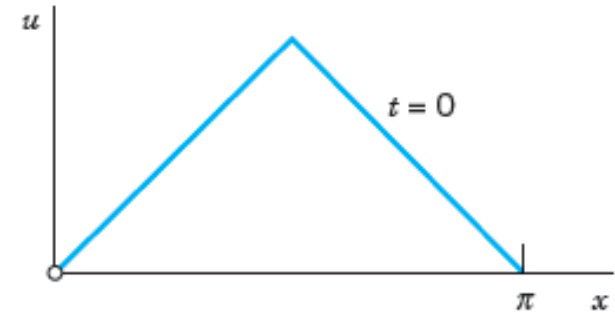
$$\frac{\partial T}{\partial t} = c^2 \frac{\partial^2 T}{\partial x^2} \quad x \in (0, L)$$

Boundary conditions

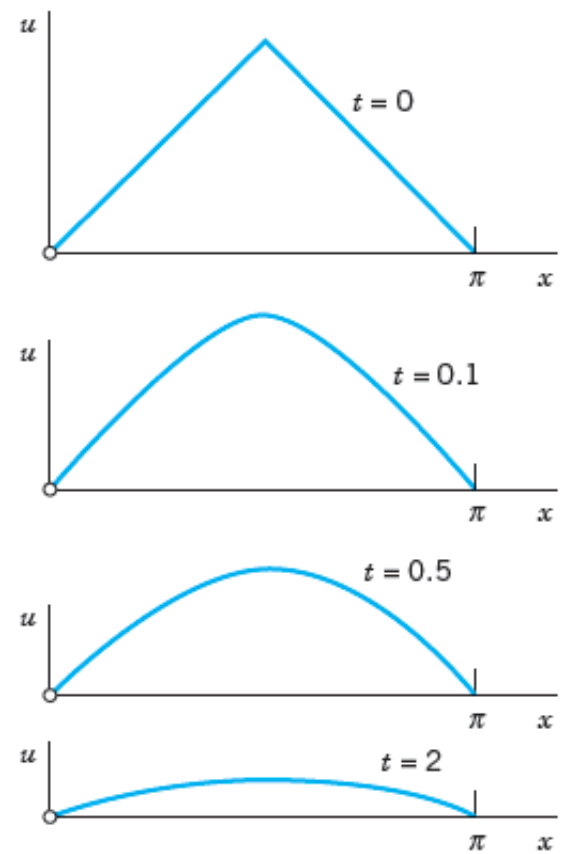
$$T(0, t) = 0, \quad T(L, t) = 0$$

Initial conditions

$$T(x, 0) = \begin{cases} x, & 0 \leq x \leq L/2 \\ (L - x), & L/2 \leq x \leq L \end{cases}$$



Initial Temperature distribution
in the bar for $L = \pi$



Decrease of Temperature with
time for $L = \pi$ and $c = 1$